Class 10: Structural Bioinformatics Pt. 1

Aarav Prasad (PID: A17440940)

Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from https://www.rcsb.org/stats/summary

```
 pdb <- read.csv("Data Export Summary.csv")
pdb
```

```
           Molecular.Type   X.ray     EM    NMR Integrative Multiple.methods
1          Protein (only) 180,758 23,111 12,813         348              229
2 Protein/Oligosaccharide  10,488  3,741     34           8               11
3              Protein/NA   9,205  6,751    287          26                8
4     Nucleic acid (only)   3,154    250  1,578           3               15
5                   Other     178     27     35           4                0
6  Oligosaccharide (only)      11      0      6           0                1
  Neutron Other   Total
1      84    32 217,375
2       1     0  14,283
3       0     0  16,277
4       3     1   5,004
5       0     0     244
6       0     4      22
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488"  "9,205"   "3,154"   "178"     "11"
```

This print out above `pdb$X.ray` is "character" not "numeric". Therefore I can't do math with it.

```
# We want to get rid / sub out commas:

x<- pdb$X.ray
tmp <- sub("," ,"", x)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do ths:

```
rm.comma <- function(x) {
  tmp <- sub("," ,"", x)
  sum(as.numeric(tmp))
}
```

```
rm.comma(pdb$EM)
```

```
[1] 33880
```

We could also use a different import function for this CSV that speaks American (i.e. dieals with commas in numbers in a comma separated value file).

```
library(readr)

pdb <-read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
pdb
```

```
# A tibble: 6 x 9
  `Molecular Type`    `X-ray`    EM   NMR Integrative `Multiple methods` Neutron
  <chr>                 <dbl> <dbl> <dbl>       <dbl>               <dbl>   <dbl>
1 Protein (only)       180758 23111 12813         348                 229      84
2 Protein/Oligosacch~   10488  3741    34           8                  11       1
3 Protein/NA             9205  6751   287          26                   8       0
4 Nucleic acid (only)    3154   250  1578           3                  15       3
5 Other                   178    27    35           4                   0       0
6 Oligosaccharide (o~      11     0     6           0                   1       0
# i 2 more variables: Other <dbl>, Total <dbl>
```

```
n.tot <- sum(pdb$Total)
n.xray <- sum(pdb$`X-ray`)
n.em <- sum(pdb$`EM`)

xray_pct <- n.xray / n.tot * 100
em_pct <- n.em / n.tot * 100
xray_pct
```

```
[1] 80.48577
```

```
em_pct
```

```
[1] 13.38046
```

Q2: What proportion of structures in the PDB are protein? How many total proteins?

```
pdb$Total[1]
```

```
[1] 217375
```

```
pdb$Total[1] / n.tot
```

```
[1] 0.8584941
```

The total number of protein sequences in Uniprot is 202,556,314

```
217375 / 202556314 * 100
```

```
[1] 0.1073158
```

> Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are ~1200 HIV-1 protease structures in the current PDB

> **Key-point**: We have a very, very small structural coverage of known proteins (0.1%). Most structures we know about (80%) come from one method (X-ray crystalography)

Visualizing PDB data with Mol-star

Main stand alone web version with all features is at https://molstar.org/viewer/

Figure 1: The HIV-protease enzyme is a homodimer with two chains

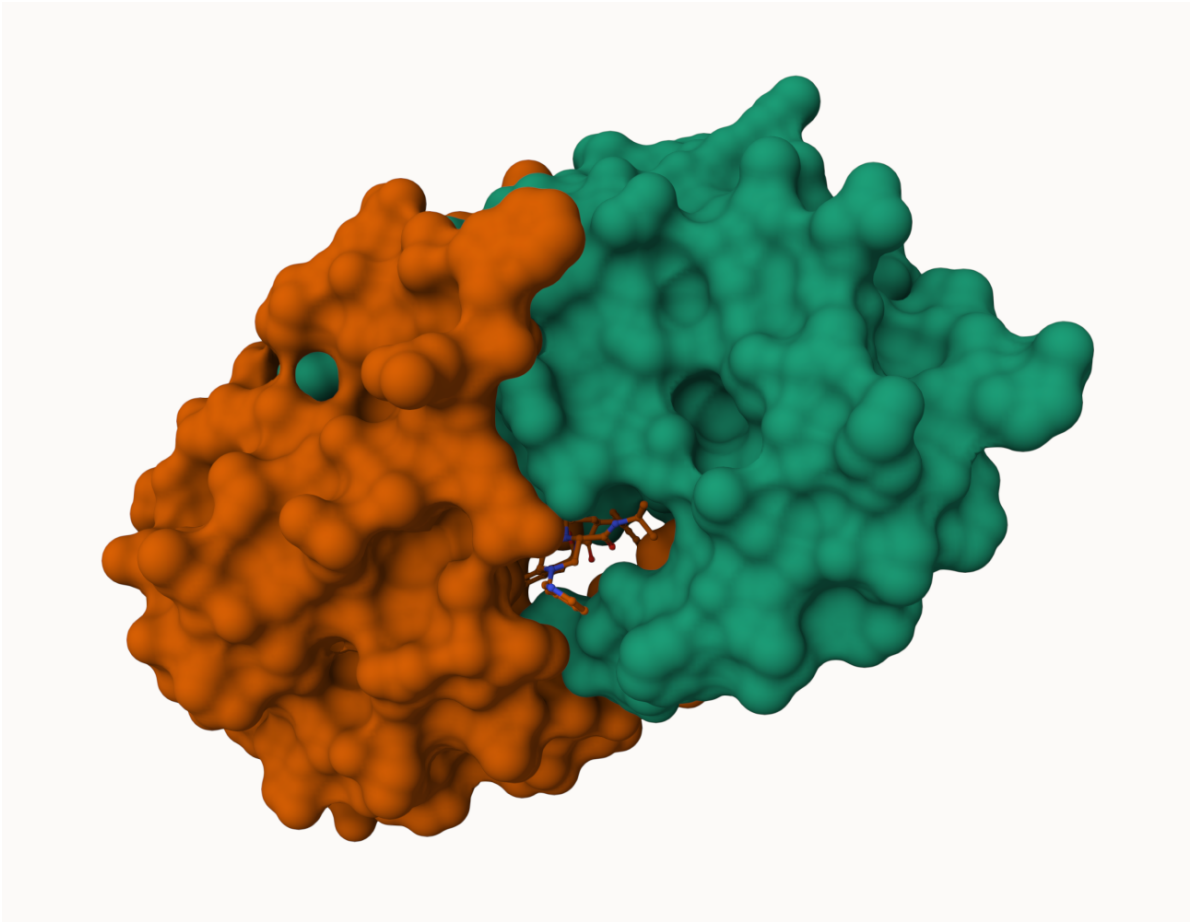


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule

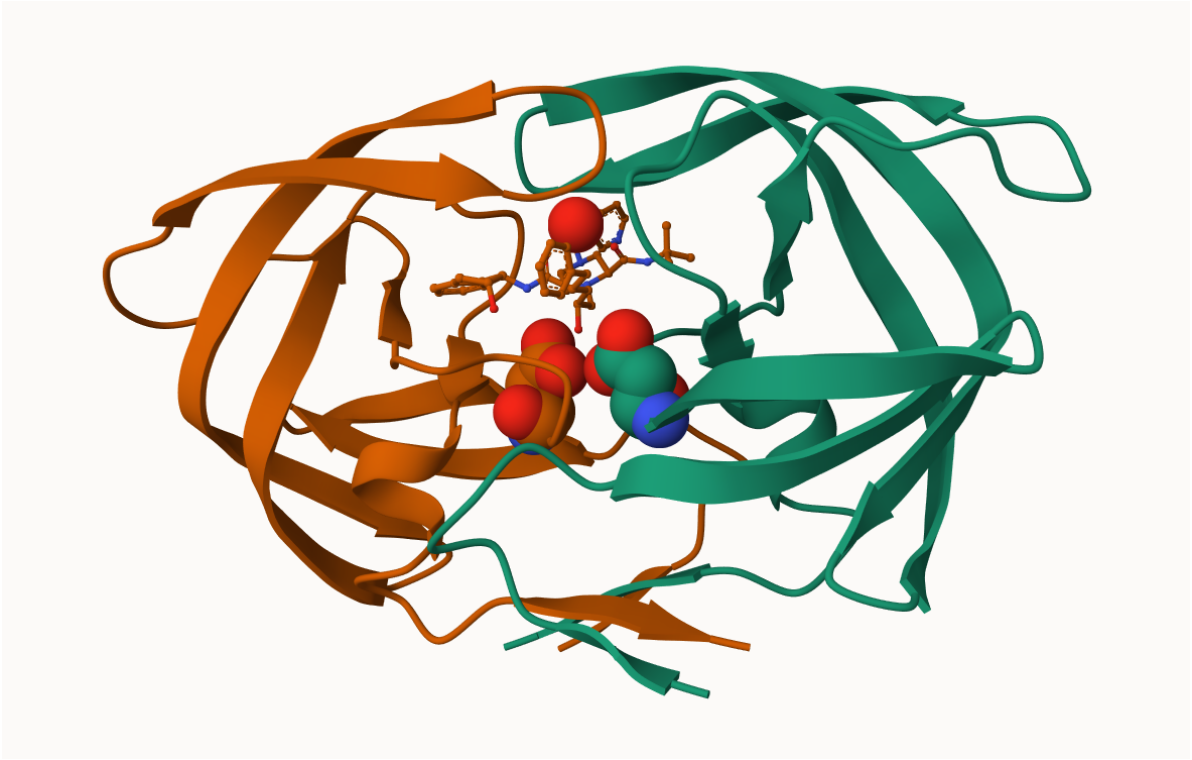


Figure 3: Spacefill dislay of catalytic ASP25 amino acids and key binding site water molecule

> Q4:Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Hydrogen atoms are usually not visible in X-ray structures because at typical resolutions (~2 Å), their electron density is too weak to detect, so only the oxygen atom of water is shown.

> Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

Residue 308

> Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.

See figures above

Getting started with the Bio3d package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
pdb
```

```
 Call:  read.pdb(file = "1hsg")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
         calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198

Q8: Name one of the two non-protein residues?

HOH

Q9: How many protein chains are in this structure?

2 chains

```
attributes(pdb)
```

```
$names
[1] "atom"   "xyz"    "seqres" "helix"  "sheet"  "calpha" "remark" "call"

$class
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

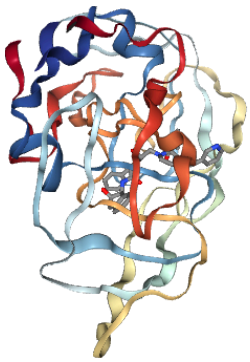
There are lots of functions that can work with these `pdb` objects:

```
head(pdbseq(pdb))
```

```
  1   2   3   4   5   6
"P" "Q" "I" "T" "L" "W"
```

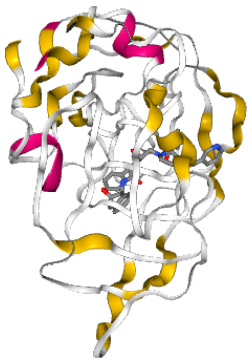
We can have a quick interactive view of any of these `pdb` objects:

```
library(bio3dview)

view.pdb(pdb)
```



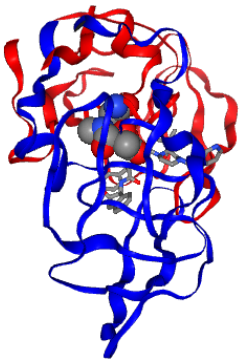
Let's try a custom view

```
view.pdb(pdb,
         colorScheme="sse",
         backgroundColor = "black")
```



Q. Create a custom view highlighting the active site ASP (`resno=25`), the two chains (in your choice of colors) and the ligand all on a custom background.

```
library(NGLVieweR)
active.site <- atom.select(pdb, resno=25)

view.pdb(pdb,
         cols = c("red", "blue"),
         highlight = active.site,
         highlight.style = "spacefill",
         backgroundColor = "pink") |>
  setRock()
```



Predict the fliexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

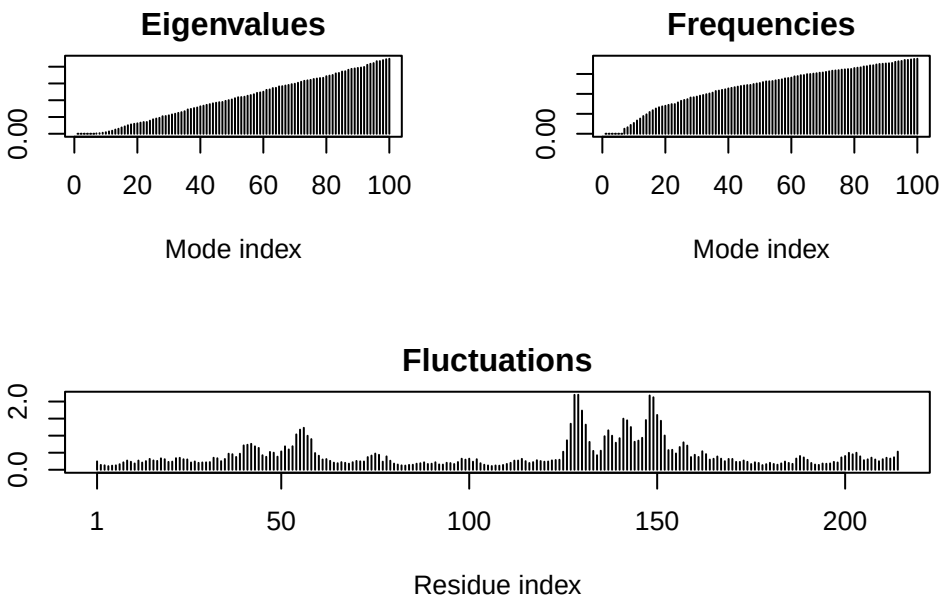
```
adk
```

```
 Call:  read.pdb(file = "6s36")

   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

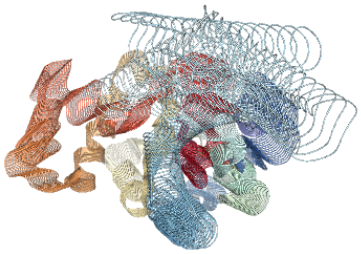
```
m <- nma(adk)
```

```
 Building Hessian...        Done in 0.014 seconds.
 Diagonalizing Hessian...   Done in 0.294 seconds.
```

```
plot(m)
```


Eigenvalues
0.00
0
20
40
60
80
100
Mode index
Frequencies
0.00
0
20
40
60
80
100
Mode index
Fluctuations
0.0
2.0
1
50
100
150
200
Residue index

```
view.nma(m)
```



Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

We will begin by first installing the packages we need for today's session.

Install packages in the R console NOT your Rmd/Quarto file

install.packages("bio3d") install.packages("NGLVieweR")

install.packages("remotes") remotes::install_github("bioboot/bio3dview")

install.packages("BiocManager") BiocManager::install("msa")

> Q10. Which of the packages above is found only on BioConductor and not CRAN?

`msa`

> Q11. Which of the above packages is not found on BioConductor or CRAN?:

`bio3dview`

> Q12:True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True

Our first step is find a sequence for this family. We will use the database ID "1ake_A" here:

```
id <- "1ake_A"

aa <- get.seq(id)
```

```
Warning in get.seq(id): Removing existing file: seqs.fasta
```

```
Fetching... Please wait. Done.
```

```
aa
```

```
             1        .         .         .         .         .         60
pdb|1AKE|A   MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
             1        .         .         .         .         .         60

            61        .         .         .         .         .         120
pdb|1AKE|A   DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
            61        .         .         .         .         .         120

           121        .         .         .         .         .         180
pdb|1AKE|A   VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
           121        .         .         .         .         .         180

           181        .         .         .   214
pdb|1AKE|A   YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG
           181        .         .         .   214

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214

Search for related sequences in the database

```
blast <- blast.pdb(aa)
```

```
 Searching ... please wait (updates every 5 seconds) RID = ZN34RPT2014
 ...
 Reporting 96 hits
```

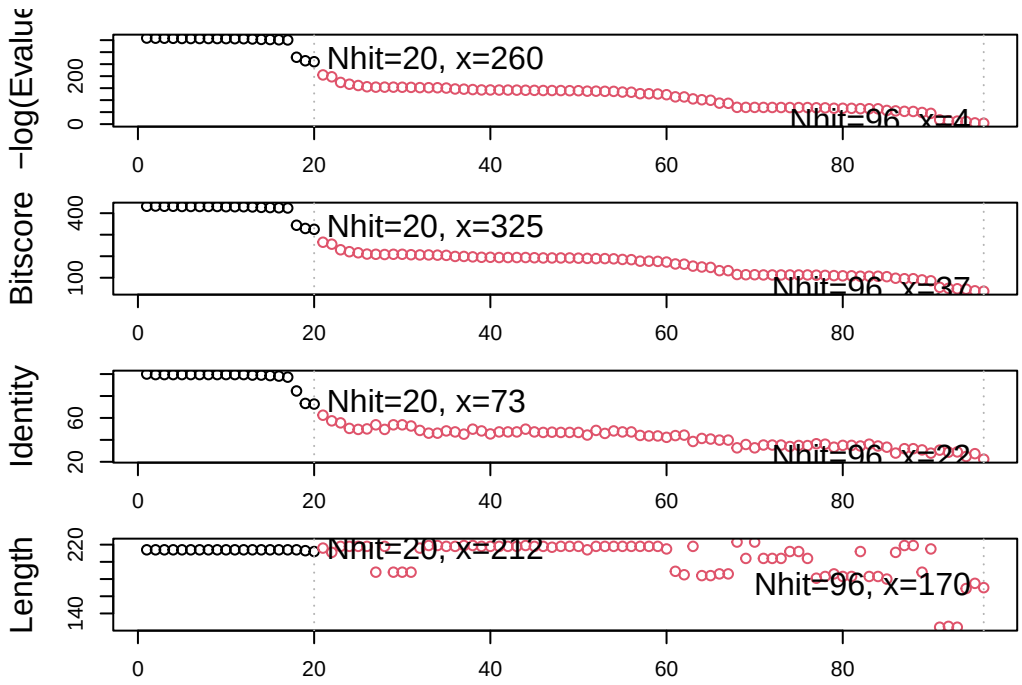
```
head(blast$hit.tbl)
```

```
        queryid subjectids identity alignmentlength mismatches gapopens q.start
```

```
1 Query_7927671     1AKE_A  100.000             214          0        0       1
2 Query_7927671     8BQF_A   99.533             214          1        0       1
3 Query_7927671     4X8M_A   99.533             214          1        0       1
4 Query_7927671     6S36_A   99.533             214          1        0       1
5 Query_7927671     9R6U_A   99.533             214          1        0       1
6 Query_7927671     9R71_A   99.533             214          1        0       1
  q.end s.start s.end    evalue bitscore positives mlog.evalue pdb.id    acc
1   214       1   214 1.81e-156      432    100.00    358.6099 1AKE_A 1AKE_A
2   214      21   234 2.97e-156      433    100.00    358.1147 8BQF_A 8BQF_A
3   214       1   214 3.25e-156      432    100.00    358.0246 4X8M_A 4X8M_A
4   214       1   214 4.76e-156      432    100.00    357.6430 6S36_A 6S36_A
5   214       1   214 1.06e-155      431     99.53    356.8424 9R6U_A 9R6U_A
6   214       1   214 1.25e-155      431     99.53    356.6775 9R71_A 9R71_A
```

```
hits <-plot(blast)
```

```
  * Possible cutoff values:    260 3
            Yielding Nhits:    20 96

  * Chosen cutoff value of:    260
            Yielding Nhits:    20
```



```
hits$pdb.id
```

```
 [1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
 [9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$pdb.id, path = "pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download


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  |=====================================================================|  90%
  |
  |=========================================================================|  95%
  |
  |=========================================================================| 100%
```

Align and supperpose all these ADK structures

```
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

```
Reading PDB files:
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/8BQF_A.pdb
```

```
pdbs/split_chain/4X8M_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/9R6U_A.pdb
pdbs/split_chain/9R71_A.pdb
pdbs/split_chain/8Q2B_A.pdb
pdbs/split_chain/8RJ9_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/4X8H_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
pdbs/split_chain/1E4Y_A.pdb
pdbs/split_chain/3X2S_A.pdb
pdbs/split_chain/6HAP_A.pdb
pdbs/split_chain/6HAM_A.pdb
pdbs/split_chain/8PVW_A.pdb
pdbs/split_chain/4K46_A.pdb
pdbs/split_chain/4NP6_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
..   PDB has ALT records, taking A only, rm.alt=TRUE
....   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

Extracting sequences

pdb/seq: 1   name: pdbs/split_chain/1AKE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbs/split_chain/8BQF_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbs/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbs/split_chain/6S36_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbs/split_chain/9R6U_A.pdb
```

```
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbs/split_chain/9R71_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbs/split_chain/8Q2B_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbs/split_chain/8RJ9_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbs/split_chain/6RZE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10   name: pdbs/split_chain/4X8H_A.pdb
pdb/seq: 11   name: pdbs/split_chain/3HPR_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12   name: pdbs/split_chain/1E4V_A.pdb
pdb/seq: 13   name: pdbs/split_chain/5EJE_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14   name: pdbs/split_chain/1E4Y_A.pdb
pdb/seq: 15   name: pdbs/split_chain/3X2S_A.pdb
pdb/seq: 16   name: pdbs/split_chain/6HAP_A.pdb
pdb/seq: 17   name: pdbs/split_chain/6HAM_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18   name: pdbs/split_chain/8PVW_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19   name: pdbs/split_chain/4K46_A.pdb
   PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 20   name: pdbs/split_chain/4NP6_A.pdb
```

```
pdbs
```

```
                                1        .        .        .         40
[Truncated_Name:1]1AKE_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:2]8BQF_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:3]4X8M_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:4]6S36_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:5]9R6U_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:6]9R71_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:7]8Q2B_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:8]8RJ9_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:9]6RZE_A.pdb    --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:10]4X8H_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:11]3HPR_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:12]1E4V_A.pdb   --MRIILLGAPVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:13]5EJE_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
```

```
[Truncated_Name:14]1E4Y_A.pdb   --MRIILLGALVAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:15]3X2S_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:16]6HAP_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:17]6HAM_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:18]8PVW_A.pdb   --MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAA
[Truncated_Name:19]4K46_A.pdb   --MRIILLGAPGAGKGTQAQFIMAKFGIPQISTGDMLRAA
[Truncated_Name:20]4NP6_A.pdb   NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTGDMLRAA
                                  ********  *********** *^**************
                                1         .         .         .         40

                               41         .         .         .         80
[Truncated_Name:1]1AKE_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:2]8BQF_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQE---
[Truncated_Name:3]4X8M_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:4]6S36_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:5]9R6U_A.pdb    VKSGSELGAQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:6]9R71_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:7]8Q2B_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:8]8RJ9_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:9]6RZE_A.pdb    VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:10]4X8H_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:11]3HPR_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:12]1E4V_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:13]5EJE_A.pdb   VKSGSELGKQAKDIMDACKLVTDELVIALVKERIAQEDCR
[Truncated_Name:14]1E4Y_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:15]3X2S_A.pdb   VKSGSELGKQAKDIMDCGKLVTDELVIALVKERIAQEDSR
[Truncated_Name:16]6HAP_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVRERICQEDSR
[Truncated_Name:17]6HAM_A.pdb   IKSGSELGKQAKDIMDAGKLVTDEIIIALVKERICQEDSR
[Truncated_Name:18]8PVW_A.pdb   VKSGSELGKQAKDIMDAGKLVTDELVIALVKERIAQEDCR
[Truncated_Name:19]4K46_A.pdb   IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA
[Truncated_Name:20]4NP6_A.pdb   IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE
                                ^* *^*** *** ^^*   **^*^^^^^*^^*** *
                               41         .         .         .         80

                               81         .         .         .         120
[Truncated_Name:1]1AKE_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:2]8BQF_A.pdb    -GFLLDGFPRTIPQADAMKEAGINVDYVIEFDVPDELIVD
[Truncated_Name:3]4X8M_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:4]6S36_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:5]9R6U_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:6]9R71_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDALIVD
[Truncated_Name:7]8Q2B_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:8]8RJ9_A.pdb    NGFLLAGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
```

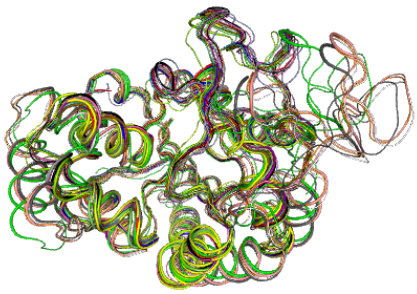
```
[Truncated_Name:9]6RZE_A.pdb    NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:10]4X8H_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:11]3HPR_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:12]1E4V_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:13]5EJE_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:14]1E4Y_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:15]3X2S_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:16]6HAP_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:17]6HAM_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:18]8PVW_A.pdb   NGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVD
[Truncated_Name:19]4K46_A.pdb   KGFLLDGFPRTIPQADGLKEVGVVVDYVIEFDVADSVIVE
[Truncated_Name:20]4NP6_A.pdb   KGFLLDGFPRTIPQADGLKEMGINVDYVIEFDVADDVIVE
                                 **** **********^^** *^ ****^**** * ^**^
                               81        .         .         .         120

                              121        .         .         .         160
[Truncated_Name:1]1AKE_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:2]8BQF_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:3]4X8M_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:4]6S36_A.pdb    KIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:5]9R6U_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:6]9R71_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:7]8Q2B_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKA
[Truncated_Name:8]8RJ9_A.pdb    RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:9]6RZE_A.pdb    AIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10]4X8H_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11]3HPR_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12]1E4V_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13]5EJE_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14]1E4Y_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15]3X2S_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16]6HAP_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17]6HAM_A.pdb   RIVGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18]8PVW_A.pdb   RILKR--GETSGRV-------------------------D
[Truncated_Name:19]4K46_A.pdb   RMAGRRAHLASGRTYHNVYNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20]4NP6_A.pdb   RMAGRRAHLPSGRTYHVVYNPPKVEGKDDVTGEDLVIRED
                                 ^  *     ***
                              121        .         .         .         160

                              161        .         .         .         200
[Truncated_Name:1]1AKE_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2]8BQF_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3]4X8M_A.pdb    DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
```

```
[Truncated_Name:4]6S36_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5]9R6U_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6]9R71_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7]8Q2B_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8]8RJ9_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9]6RZE_A.pdb    DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10]4X8H_A.pdb   DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11]3HPR_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12]1E4V_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13]5EJE_A.pdb   DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14]1E4Y_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15]3X2S_A.pdb   DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16]6HAP_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17]6HAM_A.pdb   DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18]8PVW_A.pdb   DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19]4K46_A.pdb   DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20]4NP6_A.pdb   DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
                                * ** *  **  ^*  ** ** ** *** ** * * * **
                               161        .         .         .         200

                               201        .     216
[Truncated_Name:1]1AKE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:2]8BQF_A.pdb    TKPVAEVRADLEKIL-
[Truncated_Name:3]4X8M_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:4]6S36_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:5]9R6U_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:6]9R71_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:7]8Q2B_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:8]8RJ9_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:9]6RZE_A.pdb    TKPVAEVRADLEKILG
[Truncated_Name:10]4X8H_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:11]3HPR_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:12]1E4V_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:13]5EJE_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:14]1E4Y_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:15]3X2S_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:16]6HAP_A.pdb   TKPVCEVRADLEKILG
[Truncated_Name:17]6HAM_A.pdb   TKPVCEVRADLEKILG
[Truncated_Name:18]8PVW_A.pdb   TKPVAEVRADLEKILG
[Truncated_Name:19]4K46_A.pdb   TKAVAEVSAELEKALA
[Truncated_Name:20]4NP6_A.pdb   TKQVSEVSADIAKALA
                                ** * ** *^^ * *
                               201        .     216
```

```
Call:
  pdbaln(files = files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  20 sequence rows; 216 position columns (182 non-gap, 34 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

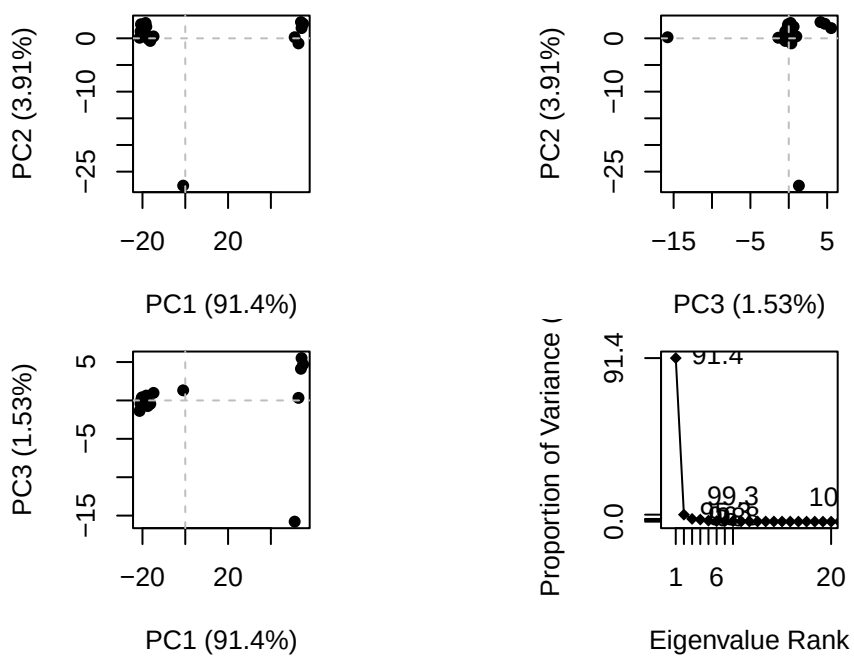
Quick interactive structural view

```
view.pdbs(pdbs)
```

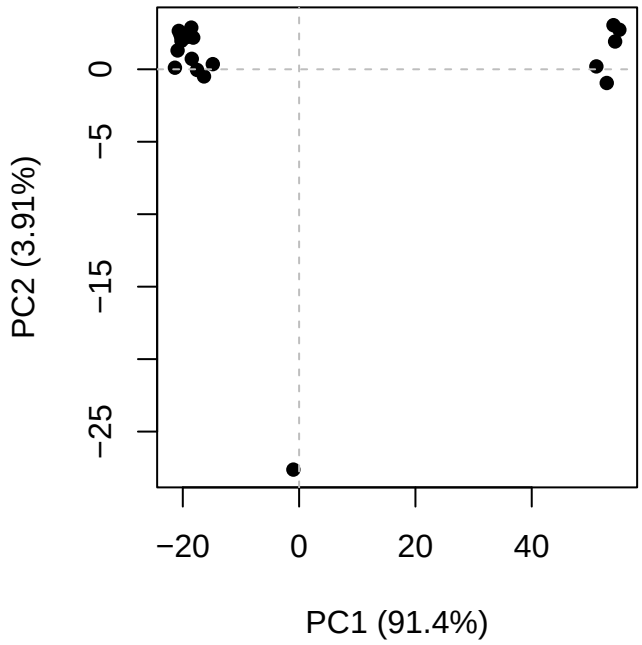


PCA of all this structural data (x, y and z atom coordinates):

```
pc <- pca(pdbs)
plot(pc)
```

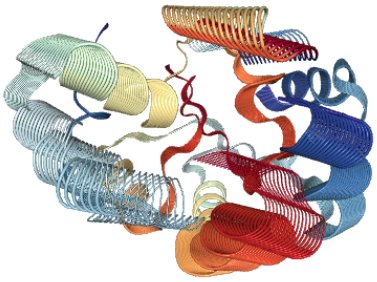



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences

```
view.pca(pc)
```



```
mktrj(pc, file = "pca.pdb")
```